

GreenNet I: Normalized Axial Green Operator

Learn one-dimensional Green kernels, not a global two-dimensional kernel



$$\mathcal{G}_{\text{unit}}[f_{\text{unit}}](t) = \int_0^1 G_{\text{unit}}(t, \eta) f_{\text{unit}}(\eta) d\eta, \quad v = \mathcal{G}_{\text{unit}}[f_{\text{unit}}].$$

Operator coefficients: Axial coefficient profiles define each local 1D operator.

Kernel coordinates (t, η) define the evaluation and source points.

GreenNet target is a local unit-interval kernel, not a global 2D Green function.